Yanke Li

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Academic Appointments

Postdoctoral Researcher in AI for Robust Clinical Prediction Modelling Oct 2026 – Present
Nuffield Department of Women's & Reproductive Health and Big Data Institute
University of Oxford, United Kingdom

- Principal investigator: Prof. Christopher Yau.
- Project: simulation-based pre-training for robust and trustworthy machine learning clinical prediction models (MRC–NIHR funded).

Education

PhD in Disease-Based Machine Learning and Statistical Modeling Sep 2022 – Sep 2026
Swiss Federal Institute of Technology Zurich (ETH Zürich), Switzerland

- Doctoral thesis: *Towards Explainable Digital Twins for Clinical Decision Support across Spinal Cord Injury and Intensive Care*.
- Thesis defended on 10 September 2026.
- Thesis advisors: Dr. Diego Paez-Granados, Prof. Robert Riener, Prof. Julia Vogt.

MSc in Statistics Sep 2019 – Nov 2021
ETH Zürich, Switzerland

- Master thesis: *Domain Adaptation with Causal Graphical Modeling*.
- Thesis supervisors: Prof. Mihaela van der Schaar (University of Cambridge) and Prof. Marloes Maathuis (ETH Zürich).
- Semester project: *Causal Discovery of Event Codes from Financial News* (funded by UBS).
- Coursework: computational statistics, machine learning, graphical models, causality, network modeling.

BSc (Hons) in Applied Mathematics Sep 2014 – Dec 2018
The Hong Kong Polytechnic University (PolyU), Hong Kong, China

- Cumulative GPA: 4.00/4.00; graduated with the Outstanding Student Award (top 1).
- Coursework: financial mathematics and computation, statistical modeling.
- Extra curriculum: data structures, databases, big data analysis, and computer vision.

Papers

Published/Accepted papers

1. **Y. Li***, R. Wang*, M. Günther and D. Paez-Granados,
GARLIC: Graph Attention-based Relational Learning of Multivariate Time Series in Intensive Care,
proceeding of *ICLR 2026*, Apr 2026.
2. **Y. Li***, T. Cui*, T. Mansi, M. Prakash and R. Liao,
BioBO: Biology-Informed Bayesian Optimization for Perturbation Design,
proceeding of *ICLR 2026*, Apr 2026.

3. J. Villines*, R. Stirnimann*, LP. Lukas, O. Taran, M. Tuci, **Y. Li**, CR. Jutzeler, JLK. Kramer, F. Geisler, D. Bourbeau, R. James Cotton+, S. C. Brüningk+, and the ASIA Engineering and Data Science Committee,
The ASIA Data Science Challenge - Predicting functional and neurological recovery from acute ISNCSCI scores.
Topics in Spinal Cord Injury Rehabilitation (TSCIR), Apr 2026.
4. **Y. Li***, H. Li* and D. Paez-Granados,
KarmaTS: A Universal Simulation Platform for Multivariate Time Series with Functional Causal Dynamics,
proceeding of *ML4H 2025*, Dec 2025.
5. T. Weikert, **Y. Li**, D. Paez-Granados and C. Awai,
Automated prediction of item-level ARAT scores from wearable sensors,
proceeding of *ICORR 2025*, May 2025.
6. **Y. Li**, A. Scheel-Sailer, R. Riener and D. Paez-Granados,
Mixed-Variable Graphical Modelling Towards Risk Prediction of Hospital-Acquired Pressure Injury in Spinal Cord Injury Individuals,
Scientific Reports, Oct 2024.
7. A. Cisnal, **Y. Li**, B. Fuchs, M. Ejtehadi, R. Riener and D. Paez-Granados,
Robust Feature Selection for Continuous BP Estimation in Multiple Populations: Towards Cuffless Ambulatory BP Monitoring,
IEEE Journal of Biomedical and Health Informatics, Jun 2024.
8. **Y. Li**, T. Hatt, I. Bica and M. van der Schaar,
Causal Domain Adaptation via Perturbed DAG Reconstruction, ETH Master Thesis (arXiv), Aug 2022.
9. J. Chen, X. Wu, **Y. Li**, Q. Li, L. Zhan and F. Chung,
A Closer Look at the Training Strategy for Modern Meta-Learning,
Proceedings of *Advances in Neural Information Processing Systems*, Dec 2020.
10. **Y. Li**,
Improvement of Embedding Channel-Wise Activation in Soft-Attention Neural Image Captioning,
Proceeding of *The 2nd International Conference on Vision, Image and Signal Processing*,
Las Vegas, USA, Aug 2018.

Under review

1. **Y. Li**, Y. Qi and D. Paez-Granados,
ICareDT: Towards Personalized Causal Digital Twin for Intensive Care,
under review at *npj Digital Medicine*, May 2026.
2. D. Paez-Granados, **Y. Li**, M. Ejtehadi, B. Fuchs and O. Gnarra
Patient Digital Twins for Caregiving Robots,
under review as a book chapter from *Springer Moonshot Consortium*, May 2026.
3. **Y. Li**, J. Stoyanov, R. Riener, R. Wettstein, S. Capossela, D. Paez-Granados and A. Bertolo,
Anaerobic Microbial Communities Influence Wound Recovery Outcome in Pressure Injuries: A Machine Learning Approach,
under review at *Scientific Reports*, Apr 2026.

In preparation

1. **Y. Li**, D. Al Jorf, A. Ighissou, A. Scheel-Sailer, R. Riener and D. Paez-Granados
Causal graphical fusion with attention steering for incomplete spinal cord injury data.
2. **Y. Li***, S. Zhong*, H. Li, R. Riener and D. Paez-Granados
Collaborative Causal Discovery with Multimodal Data in Spinal Cord Injury.
3. S. Joray, **Y. Li**, and D. Paez-Granados
GRITS: Graphical Imputation for Multivariate Time Series.

4. S. Natraj, M. Ruepp, **Y. Li**, R. Riener, and D. Paez-Granados
Monocular markerless biomechanics for clinically interpretable gait assessment in spinal cord injury.

Research experience

PhD Internship, Data Science & Artificial Intelligence Jun 2025 – Sep 2025
Johnson & Johnson Innovative Medicine

- Project: Biology-Informed Bayesian Optimization for Perturbation Design.
- Mentors: Dr. Mangal Prakash and Dr. Tianyu Cui.
- Focus: Bayesian optimization, gene perturbation, multimodal fusion, enrichment analysis.

Research Assistant, Department of Applied Mathematics Jun 2020 – Jul 2020
Hospital Authority of Hong Kong & The Hong Kong Polytechnic University

- Project: Attentive State-Space Modelling of Diabetes Progression.
- Supervisor: Prof. Daihai He.
- Focus: state-space modelling, hidden Markov modelling with attention mechanism, recurrent neural networks (LSTM), diabetes progression.

Research Assistant, Department of Applied Mathematics Feb 2019 – Aug 2019
Esquel Group & The Hong Kong Polytechnic University

- Project: Deciding Optimal Fabric Production Parameters through Yarn Data Analysis.
- Supervisors: Dr. Binyan Jiang and Prof. Daihai He.
- Focus: variable screening with LASSO, modelling and prediction using gradient boosting trees, model validation and parameter fine-tuning.

Research Trainee, Faculty of Mathematics Jun 2017 – Aug 2017
Technical University Chemnitz, Germany (IAESTE Germany)

- Project: Online Shopping Revenue Prediction and Function Approximation.
- Supervisor: Prof. Daniel Potts.
- Focus: deep learning for approximation of Fast Fourier Transform series, online shopping revenue prediction with deep learning and regression trees, model validation, and selection.

Undergraduate Research Assistant, Department of Applied Mathematics Apr 2016 – Jun 2016
The Hong Kong Polytechnic University

- Project: Mathematical Modelling of HIV Transmission.
- Supervisor: Prof. Daihai He.
- Focus: SIER compartmental model, multi-subtype agent-based modelling, visualisation of transmission network dynamics, simulation of chickenpox spread.

Teaching

Artificial Intelligence for Healthcare and Rehabilitation
ETH Zürich
Fall 2025, Teaching Assistant
Fall 2024, Teaching Assistant

Presentations & Talks

International Conference for Learning Representations (ICLR) Apr 2026
BioBO & GARLIC (Poster) Rio de Janeiro, Brazil

Machine Learning for Health Symposium (ML4H) Dec 2025
KarmaTS (Poster) San Diego, United States

International Spinal Cord Society(ISCOS) Annual Meeting Oct 2025
Mixed-Variable Graphical Modeling Towards Risk Prediction of Hospital-Acquired Pressure Injury in Spinal Cord Injury Individuals (Oral) Gothenburg, Sweden

AI for Good Global Summit May 2024
A Digital Twin Framework for Chronic Diseases (Booth) Geneva, Switzerland

European Causal Inference Meeting Apr 2023
Continual Causal Discovery and Inference for SCI Comorbidities (Poster) Oslo, Norway

Peer Review Experiences

Conferences: NeurIPS, ICLR, ICML, AISTATS, ML4H
Journals: Journal of NeuroEngineering and Rehabilitation

Mentoring & Supervision

2026, D. Al Jorf, Master Lab Practice, ETH Zürich
2025, R. Wang, Master Thesis, University of Zurich
2025, Y. Qi, Master Thesis, ETH Zürich
2025, S. Joray, Master Internship, ETH Zürich
2024, A. Ighissou, Master Thesis, Università degli Studi di Milano - Bicocca
2024, J. Koch, Semester Project, ETH Zürich
2023, H. Li, Master Internship, Technical University of Munich
2023, B. Wu, Master Thesis, Technical University of Munich

Awards and scholarships

- Machine Learning for Health 2025 Travel Award Nov 2025
- Outstanding Student Award, Faculty of Applied Science & Textiles, PolyU Feb 2018
  Awarded for the most outstanding performance during undergraduate study.
- Dean's Honours List (Top 5%), Faculty of Applied Science & Textiles, PolyU 2014–2018
- Ho & Ho Foundation Scholarships 2014–2018
  University entry scholarship for outstanding students.
- Bauhinia Cup Outstanding Entrepreneurs Association Scholarship May 2017
- Commercial Radio 50th Anniversary Scholarship Oct 2016
- Reaching Out Award, The Hong Kong Government Scholarship Jun 2015

Selected projects and competitions

ASIA Challenge: Prediction of functional recovery after traumatic spinal cord injury – 1/14
Jan 2025 – Apr 2025
Focus: predictive modeling, tabular foundation models, functional rehabilitation.

CASH Algorithm Trading Contest – Final Round Nov 2016 – Feb 2017
Focus: singular value decomposition, singular spectrum analysis, arbitrage and trading strategy.

Mathematical Contest in Modeling – Honorable Mention Feb 2016
Focus: grey relational process analysis, analytical hierarchical process, Markowitz portfolio model.

Other work experience

Data Analytic & Operations, Department of Insurance Jan 2017 – Jun 2017
Hong Kong and Shanghai Banking Corporation Limited, Hong Kong, China
Focus: detection and tracing of data anomalies, improving the online banking system, management and tracking of operational issues and production defects.

Data Analytic Intern Jun 2016 – Aug 2016
Datayes, Shanghai, China
Focus: detecting data anomalies and SQL database schema design.

Programming and Tools

Python, C++, Java, R, SAS, SQL, SPSS, Mathematica, Minitab, Microsoft Word, Microsoft Excel, Microsoft PowerPoint, TEX, LATEX.

Languages

Mandarin (native); English (proficient C2); Cantonese (fluent); German (basic A2).